

Detailed Solutions

Riemann Sums and the Definition of the Integral

Problem 1. For $f(x) = x$ on $[0, 1]$, divide $[0, 1]$ into n equal subintervals. Then

$$\Delta x = \frac{1}{n}, \quad x_i = \frac{i}{n}.$$

Using the right endpoints, the Riemann sum is

$$\sum_{i=1}^n f(x_i) \Delta x = \sum_{i=1}^n \frac{i}{n} \frac{1}{n} = \frac{1}{n^2} \sum_{i=1}^n i.$$

Since

$$\sum_{i=1}^n i = \frac{n(n+1)}{2},$$

we obtain

$$\frac{1}{n^2} \sum_{i=1}^n i = \frac{n+1}{2n}.$$

Therefore,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{i}{n}\right) \frac{1}{n} = \lim_{n \rightarrow \infty} \frac{n+1}{2n} = \frac{1}{2}.$$

Hence

$$\boxed{\int_0^1 x \, dx = \frac{1}{2}}.$$

Problem 2. For

$$\int_0^2 x^2 \, dx,$$

divide $[0, 2]$ into n equal subintervals. Then

$$\Delta x = \frac{2}{n}, \quad x_i = \frac{2i}{n}.$$

Thus the right-endpoint Riemann sum is

$$\sum_{i=1}^n \left(\frac{2i}{n}\right)^2 \frac{2}{n} = \frac{8}{n^3} \sum_{i=1}^n i^2.$$

Using

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6},$$

we get

$$\frac{8}{n^3} \frac{n(n+1)(2n+1)}{6} = \frac{4(n+1)(2n+1)}{3n^2}.$$

Therefore,

$$\int_0^2 x^2 \, dx = \lim_{n \rightarrow \infty} \frac{4(n+1)(2n+1)}{3n^2} = \frac{8}{3}.$$

Hence

$$\boxed{\int_0^2 x^2 \, dx = \frac{8}{3}}.$$

Problem 3. For

$$\int_1^3 \sqrt{x} \, dx,$$

divide $[1, 3]$ into n equal subintervals. Then

$$\Delta x = \frac{2}{n}.$$

The left endpoints are

$$x_{i-1} = 1 + \frac{2(i-1)}{n}.$$

Therefore,

$$\int_1^3 \sqrt{x} \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{1 + \frac{2(i-1)}{n}} \frac{2}{n}.$$

Equivalently, putting $j = i - 1$,

$$\boxed{\int_1^3 \sqrt{x} \, dx = \lim_{n \rightarrow \infty} \frac{2}{n} \sum_{j=0}^{n-1} \sqrt{1 + \frac{2j}{n}}.}$$

Evaluating the integral,

$$\int_1^3 \sqrt{x} \, dx = \left[\frac{2}{3} x^{3/2} \right]_1^3 = \frac{2}{3} (3\sqrt{3} - 1).$$

Thus

$$\boxed{\int_1^3 \sqrt{x} \, dx = 2\sqrt{3} - \frac{2}{3}.}$$

Problem 4. We want to evaluate

$$\int_0^1 (3x^2 + 2x + 1) \, dx.$$

The width of each subinterval is

$$\Delta x = \frac{1-0}{n} = \frac{1}{n}.$$

The right endpoint of the i -th subinterval is

$$x_i = 0 + i\Delta x = \frac{i}{n}.$$

Let

$$f(x) = 3x^2 + 2x + 1.$$

Using right endpoints,

$$\int_0^1 f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(\frac{i}{n}\right) \frac{1}{n}.$$

Therefore,

$$\begin{aligned} \int_0^1 (3x^2 + 2x + 1) \, dx &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \left[3 \left(\frac{i}{n} \right)^2 + 2 \left(\frac{i}{n} \right) + 1 \right] \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\frac{3}{n^2} \sum_{i=1}^n i^2 + \frac{2}{n} \sum_{i=1}^n i + \sum_{i=1}^n 1 \right]. \end{aligned}$$

We have

$$\sum_{i=1}^n i = \frac{n(n+1)}{2},$$

and

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

Hence,

$$\begin{aligned} \int_0^1 (3x^2 + 2x + 1) dx &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\frac{3}{n^2} \frac{n(n+1)(2n+1)}{6} + \frac{2}{n} \frac{n(n+1)}{2} + n \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{(n+1)(2n+1)}{2n^2} + \frac{n+1}{n} + 1 \right]. \end{aligned}$$

Taking the limit yields:

$$\begin{aligned} \int_0^1 (3x^2 + 2x + 1) dx &= \lim_{n \rightarrow \infty} \left[\frac{1}{2} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right) + \left(1 + \frac{1}{n} \right) + 1 \right] \\ &= \frac{1}{2}(1)(2) + 1 + 1 \\ &= 1 + 1 + 1 \\ &= \boxed{3}. \end{aligned}$$

Problem 5. Consider

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \left(1 + \frac{i}{n} \right)^3.$$

This is a Riemann sum for the function

$$f(x) = x^3$$

on the interval $[1, 2]$, since

$$\Delta x = \frac{1}{n}, \quad x_i = 1 + \frac{i}{n}.$$

Therefore,

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n} \left(1 + \frac{i}{n} \right)^3 = \int_1^2 x^3 dx.$$

Hence

$$\int_1^2 x^3 dx = \left[\frac{x^4}{4} \right]_1^2 = \frac{16 - 1}{4} = \frac{15}{4}.$$

Thus

$$\boxed{\frac{15}{4}}.$$

Riemann Integrability

Problem 6. (a) The function

$$f(x) = x^2$$

is continuous on $[0, 1]$. Every continuous function on a closed bounded interval is Riemann integrable. Hence

$$f \text{ is Riemann integrable on } [0, 1].$$

(b) The function

$$f(x) = |x - 1|$$

is continuous on $[0, 2]$. Therefore it is Riemann integrable on $[0, 2]$:

$$f \text{ is Riemann integrable on } [0, 2].$$

(c) Here

$$f(x) = \begin{cases} 1, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

The function is discontinuous only at $x = 0$. A bounded function with only finitely many points of discontinuity is Riemann integrable. Hence f is Riemann integrable. Moreover, changing the value of a function at a single point does not change its integral. Since $f(x) = 1$ except at $x = 0$,

$$\int_0^1 f(x) dx = \int_0^1 1 dx = 1.$$

Thus

$$f \text{ is Riemann integrable and } \int_0^1 f(x) dx = 1.$$

(d) Consider

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Suppose, for contradiction, that f is Riemann integrable on $[0, 1]$. Then, by the Riemann integrability criterion, for every $\varepsilon > 0$, there must exist $\delta > 0$ such that for every partition

$$P : 0 = x_0 < x_1 < \cdots < x_n = 1$$

with

$$\|P\| = \max_{1 \leq i \leq n} (x_i - x_{i-1}) < \delta,$$

and for every choice of sample points

$$t_i \in [x_{i-1}, x_i],$$

the corresponding Riemann sum

$$\sum_{i=0}^n f(t_i)(x_i - x_{i-1})$$

must be close to the value of the integral.

However, every interval $[x_{i-1}, x_i]$ contains both rational and irrational numbers.

Choose rational sample points $t_i \in [x_{i-1}, x_i]$. Then

$$f(t_i) = 1$$

for every i , and hence

$$\sum_{i=0}^n (x_i - x_{i-1}) = 1.$$

On the other hand, choose irrational sample points $s_i \in [x_{i-1}, x_i]$, Then

$$f(s_i) = 0$$

for every i , and therefore

$$\sum_{i=0}^n 0 \cdot (x_i - x_{i-1}) = 0.$$

Taking, $\varepsilon = \frac{1}{2}$, the Riemann sums cannot be made arbitrarily close to one another by taking finer partitions. This contradicts the Riemann integrability criterion. Therefore,

f is not Riemann integrable on $[0, 1]$.

Problem 7. Let

$$f(x) = \begin{cases} 1, & x \in \left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}, \\ 0, & \text{otherwise.} \end{cases}$$

Clearly,

$$0 \leq f(x) \leq 1.$$

We will show that f is Riemann integrable and that its integral is 0.

Let $\varepsilon > 0$. Choose N sufficiently large such that

$$\frac{1}{N} < \frac{\varepsilon}{2}.$$

The finitely many points

$$\frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{N}$$

can be covered by finitely many intervals whose total length is less than

$$\frac{\varepsilon}{2}.$$

Now consider any partition P that contains all the endpoints of these intervals and the point $1/N$. For any choice of sample points t_i in the subintervals of P , the corresponding Riemann sum is

$$S(f; P, \{t_i\}) = \sum_i f(t_i) \Delta x_i.$$

The function f can be nonzero only in the intervals covering

$$\frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{N}$$

or in the interval $[0, 1/N]$, since

$$\frac{1}{N+1}, \frac{1}{N+2}, \dots \in [0, 1/N].$$

Therefore,

$$S(f; P, \{t_i\}) \leq \frac{\varepsilon}{2} + \frac{1}{N} < \varepsilon.$$

Since every Riemann sum corresponding to this partition is nonnegative,

$$0 \leq S(f; P, \{t_i\}) < \varepsilon.$$

Thus, by choosing a sufficiently fine partition, every corresponding Riemann sum can be made arbitrarily close to 0. Hence f is Riemann integrable and

$$\int_0^1 f(x) dx = 0.$$

Therefore,

$f \text{ is Riemann integrable on } [0, 1] \quad \text{and} \quad \int_0^1 f(x) dx = 0.$

Problem 8. Let f be bounded on $[a, b]$, and suppose that $f = g$ except at finitely many points, where g is Riemann integrable.

Let the exceptional points be

$$x_1, \dots, x_N.$$

Since f and g are bounded, there exists $M > 0$ such that

$$|f(x)| \leq M, \quad |g(x)| \leq M$$

for all $x \in [a, b]$.

We show that f is Riemann integrable.

Let $\varepsilon > 0$. Choose disjoint intervals

$$I_1, \dots, I_N$$

containing $x_1, \dots, x_N$, respectively, such that

$$\sum_{j=1}^N |I_j| < \frac{\varepsilon}{4M}.$$

Let P_0 be a partition of $[a, b]$ such that every Riemann sum for g corresponding to P_0 differs from

$$\int_a^b g(x) dx$$

by less than $\varepsilon/2$.

Now refine P_0 by adding all the endpoints of the intervals $I_1, \dots, I_N$. Denote the resulting partition by P .

Consider any two choices of sample points

$$t_i, s_i \in [x_{i-1}, x_i]$$

for the partition P . We compare the corresponding Riemann sums

$$S(f; P, \{t_i\}) = \sum_i f(t_i) \Delta x_i$$

and

$$S(f; P, \{s_i\}) = \sum_i f(s_i) \Delta x_i.$$

For every subinterval of P lying outside

$$I_1 \cup \dots \cup I_N,$$

we have $f = g$ throughout that subinterval. Hence the contributions from such subintervals agree with the corresponding contributions for g .

The only possible difference occurs in the intervals $I_1, \dots, I_N$. Since

$$|f| \leq M,$$

the difference between the two contributions from these intervals is at most

$$2M \sum_{j=1}^N |I_j| < 2M \cdot \frac{\varepsilon}{4M} = \frac{\varepsilon}{2}.$$

Thus, all Riemann sums of f corresponding to P differ from the corresponding Riemann sums of g by less than $\varepsilon/2$. Since g is Riemann integrable, the Riemann sums of g can be made arbitrarily close to

$$\int_a^b g(x) dx.$$

Consequently, the Riemann sums of f can be made arbitrarily close to the same number. Therefore f is Riemann integrable.

Moreover, because the two functions differ only at finitely many points, the contribution of those points to any Riemann sum can be made arbitrarily small. Hence their Riemann integrals are equal:

$$\boxed{\int_a^b f(x) dx = \int_a^b g(x) dx.}$$

Problem 9. Since f is Riemann integrable, f is bounded and for every $\varepsilon > 0$, there exists a partition P such that

$$U(f, P) - L(f, P) < \varepsilon.$$

Let

$$M_i(f) = \max_{[x_{i-1}, x_i]} f(x), \quad m_i(f) = \min_{[x_{i-1}, x_i]} f(x).$$

For $x, y \in [x_{i-1}, x_i]$, we have

$$||f(x)| - |f(y)|| \leq |f(x) - f(y)|.$$

Therefore,

$$\max_{[x_{i-1}, x_i]} |f| - \min_{[x_{i-1}, x_i]} |f| \leq \max_{[x_{i-1}, x_i]} f - \min_{[x_{i-1}, x_i]} f.$$

Multiplying by Δx_i and adding over all the subintervals,

$$U(|f|, P) - L(|f|, P) \leq U(f, P) - L(f, P) < \varepsilon.$$

Hence, for every $\varepsilon > 0$, there exists a partition P such that

$$U(|f|, P) - L(|f|, P) < \varepsilon.$$

By the Riemann integrability criterion, $|f|$ is Riemann integrable on $[a, b]$.

Therefore,

$$\boxed{|f| \text{ is Riemann integrable on } [a, b].}$$

Properties of the Riemann Integral

Problem 10. (a) Let c be a constant. For a partition

$$P = \{x_0, x_1, \dots, x_n\},$$

the Riemann sums for cf are

$$\sum_{i=1}^n cf(t_i)\Delta x_i = c \sum_{i=1}^n f(t_i)\Delta x_i.$$

Taking limits gives

$$\boxed{\int_a^b cf(x) dx = c \int_a^b f(x) dx}.$$

Hence cf is Riemann integrable.

(b) Consider a forward partition $P = \{x_0, x_1, x_2, \dots, x_n\}$ from a to b where:

$$a = x_0 < x_1 < x_2 < \dots < x_n = b$$

with subinterval widths $\Delta x_i = x_i - x_{i-1} > 0$ and sample points $t_i \in [x_{i-1}, x_i]$. The corresponding Riemann sum is:

$$S(f, P) = \sum_{i=1}^n f(t_i)\Delta x_i = \sum_{i=1}^n f(t_i)(x_i - x_{i-1})$$

Now, consider the reversed partition P' from b to a defined by points $y_i = x_{n-i}$ for $i = 0, 1, \dots, n$, such that:

$$b = y_0 > y_1 > y_2 > \dots > y_n = a$$

The directed width of each subinterval in P' is:

$$\Delta y_i = y_i - y_{i-1} = x_{n-i} - x_{n-i+1} = -(x_{n-i+1} - x_{n-i}) = -\Delta x_{n-i+1}$$

Using the corresponding sample points $t_{n-i+1} \in [x_{n-i}, x_{n-i+1}]$, the Riemann sum for the integral from b to a is:

$$S(f, P') = \sum_{i=1}^n f(t_{n-i+1}) \Delta y_i = \sum_{i=1}^n f(t_{n-i+1}) (-\Delta x_{n-i+1}) = - \sum_{i=1}^n f(t_{n-i+1}) \Delta x_{n-i+1}$$

Re-indexing the summation using $k = n - i + 1$ yields:

$$S(f, P') = - \sum_{k=1}^n f(t_k) \Delta x_k = -S(f, P)$$

Taking the limit as the mesh size $\|P\| \rightarrow 0$ (and thus $\|P'\| \rightarrow 0$):

$$\int_b^a f(x) dx = \lim_{\|P'\| \rightarrow 0} S(f, P') = \lim_{\|P\| \rightarrow 0} (-S(f, P)) = - \int_a^b f(x) dx$$

Rearranging terms gives the required result:

$$\int_a^b f(x) dx = - \int_b^a f(x) dx \quad \blacksquare$$

(c) For any partition and sample points,

$$\sum_{i=1}^n (f(t_i) + g(t_i)) \Delta x_i = \sum_{i=1}^n f(t_i) \Delta x_i + \sum_{i=1}^n g(t_i) \Delta x_i.$$

Taking limits gives

$$\boxed{\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx}.$$

Thus $f + g$ is Riemann integrable.

(d) Suppose

$$f(x) \leq g(x) \quad \text{for all } x \in [a, b].$$

Then

$$g(x) - f(x) \geq 0.$$

By the nonnegativity property of the integral,

$$\int_a^b (g(x) - f(x)) dx \geq 0.$$

Using linearity,

$$\int_a^b g(x) dx - \int_a^b f(x) dx \geq 0.$$

Therefore,

$$\boxed{\int_a^b f(x) dx \leq \int_a^b g(x) dx}.$$

(e) From

$$m \leq f(x) \leq M,$$

we obtain

$$m \leq f(x) \leq M.$$

Integrating throughout $[a, b]$ gives

$$\int_a^b m \, dx \leq \int_a^b f(x) \, dx \leq \int_a^b M \, dx.$$

Hence

$$m(b-a) \leq \int_a^b f(x) \, dx \leq M(b-a).$$

Thus

$$\boxed{m(b-a) \leq \int_a^b f(x) \, dx \leq M(b-a)}.$$

(f) Since

$$-f(x) \leq |f(x)| \leq f(x)$$

is not generally true, we instead use

$$-|f(x)| \leq f(x) \leq |f(x)|.$$

By monotonicity,

$$-\int_a^b |f(x)| \, dx \leq \int_a^b f(x) \, dx \leq \int_a^b |f(x)| \, dx.$$

Therefore,

$$\boxed{\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx}.$$

Problem 11. (a) Suppose f is odd. Then

$$f(-x) = -f(x).$$

Using a change of variable $x = -u$,

$$\int_{-a}^0 f(x) \, dx = \int_a^0 f(-u)(-du) = \int_0^a f(-u) \, du.$$

Since f is odd,

$$\int_{-a}^0 f(x) \, dx = - \int_0^a f(u) \, du.$$

Hence

$$\int_{-a}^a f(x) \, dx = - \int_0^a f(x) \, dx + \int_0^a f(x) \, dx = 0.$$

Thus

$$\boxed{\int_{-a}^a f(x) \, dx = 0}.$$

(b) Suppose f is even. Then

$$f(-x) = f(x).$$

As above,

$$\int_{-a}^0 f(x) dx = \int_0^a f(-u) du = \int_0^a f(u) du.$$

Therefore,

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx.$$

Thus

$$\boxed{\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx}.$$

Average Value of a Function

Problem 12. (a) For

$$f(x) = x^2, \quad [a, b] = [0, 3],$$

the average value is

$$\text{av}(f) = \frac{1}{3-0} \int_0^3 x^2 dx.$$

Therefore,

$$\text{av}(f) = \frac{1}{3} \left[\frac{x^3}{3} \right]_0^3 = \frac{1}{3} \cdot 9 = 3.$$

Hence

$$\boxed{3}.$$

(b) For

$$f(x) = \sin x, \quad [a, b] = [0, \pi],$$

we have

$$\text{av}(f) = \frac{1}{\pi} \int_0^\pi \sin x dx.$$

Thus

$$\text{av}(f) = \frac{1}{\pi} [-\cos x]_0^\pi = \frac{1}{\pi} (1 + 1) = \frac{2}{\pi}.$$

Hence

$$\boxed{\frac{2}{\pi}}.$$

Problem 13. Since

$$|f(x)| \leq 3,$$

we have

$$\left| \int_0^2 f(x) dx \right| \leq \int_0^2 |f(x)| dx \leq \int_0^2 3 dx.$$

Therefore,

$$\left| \int_0^2 f(x) dx \right| \leq 3(2) = 6.$$

Hence

$$\boxed{\left| \int_0^2 f(x) dx \right| \leq 6}.$$

Areas Between Curves

Problem 14. (a) We have

$$f(x) = x^2 - 4x + 3 = (x - 1)(x - 3).$$

The zeros are $x = 1$ and $x = 3$. On $[0, 1]$, $f(x) \geq 0$, while on $[1, 3]$, $f(x) \leq 0$. Hence the total area is

$$A = \int_0^1 (x^2 - 4x + 3) dx - \int_1^3 (x^2 - 4x + 3) dx.$$

An antiderivative is

$$F(x) = \frac{x^3}{3} - 2x^2 + 3x.$$

Therefore,

$$\int_0^1 f(x) dx = \frac{4}{3},$$

and

$$- \int_1^3 f(x) dx = \frac{4}{3}.$$

Thus

$$\boxed{A = \frac{8}{3}}.$$

(b) The curve

$$y = 1 - \frac{x^2}{4}$$

is nonnegative on $[-2, 2]$ and negative on $[2, 3]$.

Therefore,

$$A = \int_{-2}^2 \left(1 - \frac{x^2}{4}\right) dx + \int_2^3 \left(\frac{x^2}{4} - 1\right) dx.$$

For the first integral, using symmetry,

$$\int_{-2}^2 \left(1 - \frac{x^2}{4}\right) dx = 2 \int_0^2 \left(1 - \frac{x^2}{4}\right) dx = \frac{8}{3}.$$

Also,

$$\int_2^3 \left(\frac{x^2}{4} - 1\right) dx = \frac{5}{12}.$$

Hence

$$A = \frac{8}{3} + \frac{5}{12} = \boxed{\frac{13}{4}}.$$

(c) We have

$$f(x) = 5 - 5x^{2/3}.$$

The curve meets the x -axis when

$$5 - 5x^{2/3} = 0 \implies x^{2/3} = 1 \implies x = \pm 1.$$

Thus $f(x) \geq 0$ on $[-1, 1]$ and $f(x) \leq 0$ on $[1, 8]$.

Therefore,

$$A = \int_{-1}^1 (5 - 5x^{2/3}) dx + \int_1^8 (5x^{2/3} - 5) dx.$$

Using

$$\int x^{2/3} dx = \frac{3}{5} x^{5/3},$$

the first area is

$$\int_{-1}^1 (5 - 5x^{2/3}) dx = 4.$$

The second area is

$$\int_1^8 (5x^{2/3} - 5) dx = 58.$$

Hence

$$\boxed{A = 62}.$$

Problem 15. (a) The curves

$$y = x, \quad y = \frac{1}{x^2}$$

intersect when

$$x = \frac{1}{x^2} \implies x^3 = 1 \implies x = 1.$$

Thus the required area, from $x = 1$ to $x = 2$, is

$$A = \int_1^2 \left(x - \frac{1}{x^2} \right) dx.$$

Since

$$\int \left(x - \frac{1}{x^2} \right) dx = \frac{x^2}{2} + \frac{1}{x},$$

we obtain

$$A = \left[\frac{x^2}{2} + \frac{1}{x} \right]_1^2 = \frac{5}{2} - \frac{3}{2} = \boxed{1}.$$

(b) The curves intersect when

$$x = \frac{1}{\sqrt{x}} \implies x^{3/2} = 1 \implies x = 1.$$

For $1 \leq x \leq 2$, $y = x$ is above $y = x^{-1/2}$. Hence

$$A = \int_1^2 \left(x - \frac{1}{\sqrt{x}} \right) dx.$$

Therefore,

$$A = \left[\frac{x^2}{2} - 2\sqrt{x} \right]_1^2.$$

Thus

$$A = (2 - 2\sqrt{2}) - \left(\frac{1}{2} - 2 \right) = \boxed{\frac{7}{2} - 2\sqrt{2}}.$$

(c) From

$$\sqrt{x} + \sqrt{y} = 1,$$

we obtain

$$\sqrt{y} = 1 - \sqrt{x},$$

so

$$y = (1 - \sqrt{x})^2.$$

The region lies between $x = 0$ and $x = 1$. Hence

$$A = \int_0^1 (1 - \sqrt{x})^2 dx.$$

Expanding,

$$A = \int_0^1 (1 - 2\sqrt{x} + x) dx.$$

Therefore,

$$A = \left[x - \frac{4}{3}x^{3/2} + \frac{x^2}{2} \right]_0^1 = 1 - \frac{4}{3} + \frac{1}{2} = \boxed{\frac{1}{6}}.$$

Problem 16. The lower boundary is $y = 1$. The upper boundary is

$$y = \sqrt{x}$$

for the left part and

$$y = 6 - x$$

for the right part.

The intersection of $y = \sqrt{x}$ and $y = 1$ gives

$$\sqrt{x} = 1 \implies x = 1.$$

The intersection of $y = 6 - x$ and $y = 1$ gives

$$6 - x = 1 \implies x = 5.$$

The two upper curves intersect at

$$\sqrt{x} = 6 - x.$$

Let $t = \sqrt{x}$. Then

$$t = 6 - t^2,$$

so

$$t^2 + t - 6 = 0 \implies t = 2,$$

and hence

$$x = 4.$$

Thus

$$A = \int_1^4 (\sqrt{x} - 1) dx + \int_4^5 ((6 - x) - 1) dx.$$

For the first integral,

$$\int_1^4 (\sqrt{x} - 1) dx = \left[\frac{2}{3}x^{3/2} - x \right]_1^4 = \frac{5}{3}.$$

For the second,

$$\int_4^5 (5 - x) dx = \frac{1}{2}.$$

Hence

$$A = \frac{5}{3} + \frac{1}{2} = \boxed{\frac{13}{6}}.$$

Problem 17. We need the total area between

$$y = \sin x \quad \text{and} \quad y = \cos x$$

on

$$0 \leq x \leq \frac{3\pi}{2}.$$

The curves intersect when

$$\sin x = \cos x \implies \tan x = 1.$$

Thus, in the given interval, the intersection points are

$$x = \frac{\pi}{4}, \quad x = \frac{5\pi}{4}.$$

Therefore,

$$A = \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{5\pi/4} (\sin x - \cos x) dx + \int_{5\pi/4}^{3\pi/2} (\cos x - \sin x) dx.$$

Evaluating,

$$\int_0^{\pi/4} (\cos x - \sin x) dx = \sqrt{2} - 1,$$

$$\int_{\pi/4}^{5\pi/4} (\sin x - \cos x) dx = 2\sqrt{2},$$

and

$$\int_{5\pi/4}^{3\pi/2} (\cos x - \sin x) dx = \sqrt{2} - 1.$$

Hence

$$A = (\sqrt{2} - 1) + 2\sqrt{2} + (\sqrt{2} - 1) = \boxed{4\sqrt{2} - 2}.$$

Problem 18. For $1 \leq x \leq e$,

$$\ln x \geq 0.$$

Thus the required area is

$$A = \int_1^e \frac{2 \ln x}{x} dx.$$

Since

$$\int \frac{\ln x}{x} dx = \frac{1}{2}(\ln x)^2,$$

we get

$$A = [(\ln x)^2]_1^e = 1 - 0 = \boxed{1}.$$

Volumes of Solids with Known Cross-Sections

Problem 19. (a) The diagonal of each square extends from

$$y = -\sqrt{x}$$

to

$$y = \sqrt{x}.$$

Therefore the diagonal has length

$$d = 2\sqrt{x}.$$

If s is the side length of the square, then

$$d = s\sqrt{2}.$$

Hence

$$s = \frac{2\sqrt{x}}{\sqrt{2}} = \sqrt{2x}.$$

The cross-sectional area is therefore

$$A(x) = s^2 = 2x.$$

Thus

$$V = \int_0^4 A(x) dx = \int_0^4 2x dx = [x^2]_0^4 = \boxed{16}.$$

(b) The base of each square has length

$$2\sqrt{1-x^2}.$$

This is the side length of the square, so

$$A(x) = \left(2\sqrt{1-x^2}\right)^2 = 4(1-x^2).$$

Therefore,

$$V = \int_{-1}^1 4(1-x^2) dx.$$

Using symmetry,

$$V = 8 \int_0^1 (1-x^2) dx = 8 \left[x - \frac{x^3}{3} \right]_0^1 = 8 \left(1 - \frac{1}{3} \right).$$

Hence

$$\boxed{V = \frac{16}{3}}.$$

(c) The diameter of each circular disk is

$$d = \sec x - \tan x.$$

Hence its radius is

$$r = \frac{\sec x - \tan x}{2}.$$

Therefore,

$$A(x) = \pi r^2 = \frac{\pi}{4}(\sec x - \tan x)^2.$$

Thus

$$V = \frac{\pi}{4} \int_{-\pi/3}^{\pi/3} (\sec x - \tan x)^2 dx.$$

Using

$$(\sec x - \tan x)^2 = \sec^2 x + \tan^2 x - 2 \sec x \tan x,$$

an antiderivative is

$$2 \tan x - 2 \ln(\sec x + \tan x) - x.$$

Hence

$$V = \frac{\pi}{4} [2 \tan x - 2 \ln(\sec x + \tan x) - x]_{-\pi/3}^{\pi/3}.$$

This gives

$$\boxed{V = \pi\sqrt{3} - \frac{\pi^2}{6}}.$$

For square cross-sections, the side length is

$$s = \sec x - \tan x.$$

Thus

$$A(x) = (\sec x - \tan x)^2,$$

and

$$V = \int_{-\pi/3}^{\pi/3} (\sec x - \tan x)^2 dx.$$

Using the same antiderivative,

$$\boxed{V = 4\sqrt{3} - \frac{2\pi}{3}}.$$

Volumes of Revolution

Problem 20. (a) The region is bounded by

$$y = x^2, \quad y = 0, \quad x = 2.$$

Revolving about the x -axis gives disks of radius

$$R(x) = x^2.$$

Hence

$$V = \pi \int_0^2 R(x)^2 dx = \pi \int_0^2 x^4 dx.$$

Therefore,

$$V = \pi \left[\frac{x^5}{5} \right]_0^2 = \frac{32\pi}{5}.$$

Thus

$$\boxed{V = \frac{32\pi}{5}}.$$

(b) The radius of a disk at x is

$$R(x) = \sqrt{\cos x}.$$

Therefore,

$$V = \pi \int_0^{\pi/2} R(x)^2 dx = \pi \int_0^{\pi/2} \cos x dx.$$

Hence

$$V = \pi [\sin x]_0^{\pi/2} = \boxed{\pi}.$$

(c) For

$$y = \sqrt{x \cot x},$$

the radius of the disk is

$$R(x) = \sqrt{x \cot x}.$$

Therefore,

$$R(x)^2 = x \cot x.$$

Hence

$$V = \pi \int_{\pi/6}^{\pi/2} x \cot x dx.$$

Thus

$$\boxed{V = \pi \int_{\pi/6}^{\pi/2} x \cot x dx}.$$

Problem 21. (a) The equation

$$x = \sqrt{2y}$$

is equivalent to

$$y = \frac{x^2}{2}.$$

However, $\sqrt{2y}$ is real only when

$$y \geq 0.$$

Thus the stated region with

$$-1 \leq y \leq 1$$

is not a real-valued region for $-1 \leq y < 0$.

If the intended lower bound is $y = 0$, then revolving

$$x = \sqrt{2y}$$

about the y -axis gives disks of radius

$$R(y) = \sqrt{2y}.$$

Hence

$$V = \pi \int_0^1 R(y)^2 dy = \pi \int_0^1 2y dy = \pi.$$

Thus, with the corrected interval $0 \leq y \leq 1$,

$$\boxed{V = \pi}.$$

(b) For

$$x = \frac{2}{\sqrt{y+1}}, \quad 0 \leq y \leq 3,$$

the radius of a disk obtained by revolving about the y -axis is

$$R(y) = \frac{2}{\sqrt{y+1}}.$$

Hence

$$V = \pi \int_0^3 \left(\frac{2}{\sqrt{y+1}} \right)^2 dy = 4\pi \int_0^3 \frac{1}{y+1} dy.$$

Therefore,

$$V = 4\pi [\ln(y+1)]_0^3 = 4\pi \ln 4 = \boxed{8\pi \ln 2}.$$

(c) For

$$x = \sqrt{\frac{2y}{y^2+1}}, \quad 0 \leq y \leq 1,$$

the radius is

$$R(y) = \sqrt{\frac{2y}{y^2+1}}.$$

Therefore,

$$V = \pi \int_0^1 \frac{2y}{y^2+1} dy.$$

Let

$$u = y^2 + 1, \quad du = 2y dy.$$

Then

$$V = \pi \int_1^2 \frac{1}{u} du = \pi [\ln u]_1^2 = \boxed{\pi \ln 2}.$$